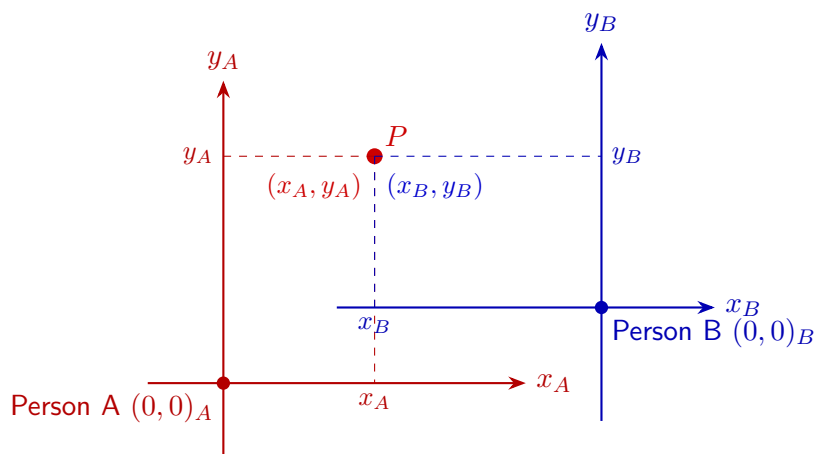


Chapter 3

2D Vectors

3.1 Vector Notation

Recall that a coordinate plane represents positions using an ordered pair of numbers, (x, y) , indicating the horizontal and vertical displacement of an object from the origin. If two observers construct separate maps using their respective locations as origin points, their coordinate representations of the same physical point will differ as shown below:



Although both observers obtain a correct local illustration for point P , coordinate values shift whenever the frame of reference moves. Physical displacement has a fundamental impact on how we perceive positions.

To describe displacement independently of reference frames, we must incorporate both **direction** and **distance**. We introduce the mathematical concept of a **vector** to achieve this goal.

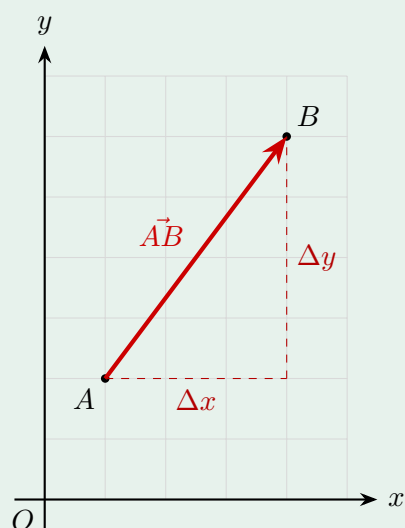
Definition 3.1.1: Intuition of Vector

A **vector** is a mathematical quantity that possesses both **magnitude** (length or size) and **direction**. Geometrically, a vector from an initial point A to a terminal point B is represented by a directed line segment (or arrow) from A to B , denoted as $\vec{AB}$ (or $\overrightarrow{AB}$).

For example, we can draw a vector on a map to illustrate how position changes when an object moves from point A to point B .

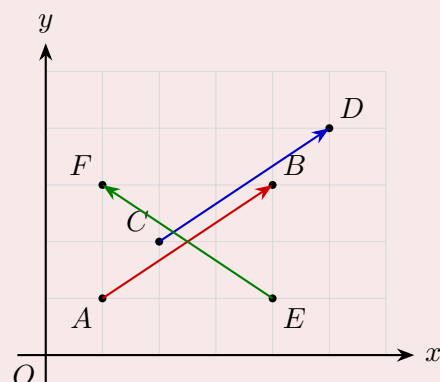
Example 3.1.1: Vector as Position Change

A particle moves on a coordinate plane from point $A(1, 2)$ to point $B(4, 6)$.

**Exercise 3.1.1: Vector as Position Change**

The figure shows a grid with several points A, B, C, D, E , and F .

- Trace the path on the grid and describe the horizontal shift (Δx) and vertical shift (Δy) represented by the vector $\vec{AB}$.
- Based on the grid, determine which two vectors represent the exact same displacement in terms of direction and distance. Explain briefly using grid units.
- Draw a directed line segment on the grid starting from point F to represent a displacement equal in magnitude but opposite in direction to $\vec{AB}$. Label its endpoint as G .



If we consider displacement from an arbitrary origin defined within the system, the vector explicitly points to the coordinate location. This idea provides a framework for **decentralization**.

Remark

To distinguish line segments from vectors, an arrow is placed above the point notation to specify initial and terminal points. This notation explicitly incorporates orientation:

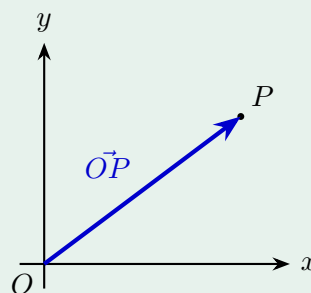
$$OP = PO \quad \text{whereas} \quad \vec{OP} \neq \vec{PO}$$

One primary advantage of vectors is that they allow us to replace static coordinates with dynamic position vectors.

Example 3.1.2: Representing Coordinates Using Vectors

In the figure, the position of point P relative to origin O is uniquely specified by the vector $\vec{OP}$.

The vector $\vec{OP}$ is called the **position vector** of point P relative to origin O .

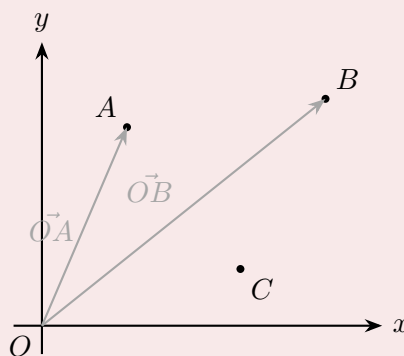

Exercise 3.1.2: Representing Coordinates Using Vectors

In the figure, O is the origin, and A , B , and C are three distinct points on the plane.

- Construct the position vector $\vec{OC}$ corresponding to point C .
- Draw the directed line segment representing the displacement vector $\vec{AB}$.
- Using the head-to-tail path from O to B via A , complete the relation:

$$\vec{OA} + \vec{AB} = \underline{\hspace{2cm}}$$

Hence, write down $\vec{AB}$ in terms of the position vectors $\vec{OA}$ and $\vec{OB}$.



Although vectors share structural similarities with coordinate pairs, vector operations provide significantly richer tools. Our objective is to abstract coordinate systems into coordinate-free vector algebra.

Follow-up Exercises
Level 1 (Basic)

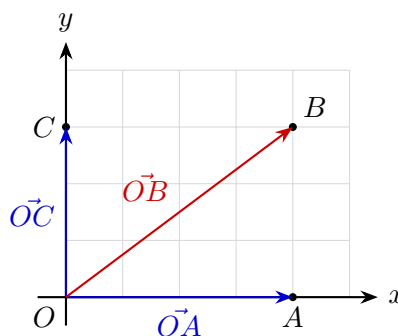
- The rectangle $OABC$ is drawn on a grid with origin O , as shown below.

- State the vector representing the displacement from C to B in terms of $\vec{OA}$.
- State the vector representing the displacement from A to B in terms of $\vec{OC}$.
- Using the path $O \rightarrow A \rightarrow B$, complete the relation:

$$\vec{OA} + \vec{AB} = \underline{\hspace{2cm}}$$

- Using the path $O \rightarrow C \rightarrow B$, complete the relation:

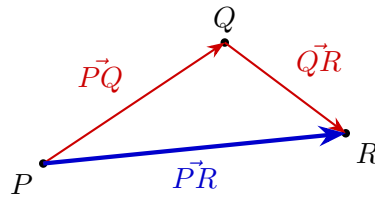
$$\vec{OC} + \vec{CB} = \underline{\hspace{2cm}}$$



- The figure shows three points P , Q , and R connected by directed line segments.

- (a) Complete the vector addition statement representing net displacement from P to R :

$$\vec{PQ} + \underline{\hspace{2cm}} = \vec{PR}$$

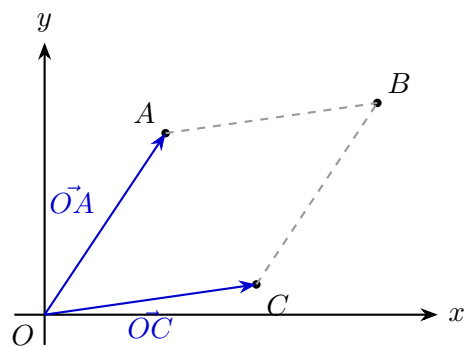


- (b) Express $\vec{QR}$ in terms of $\vec{PQ}$ and $\vec{PR}$.
- (c) Is $\vec{RP}$ equal to $\vec{PR}$? Explain briefly using vector direction.

Level 2 (Advanced)

3. In the figure, $OABC$ is a parallelogram on the coordinate plane, where O is the origin. The position vectors of vertices A and C relative to O are $\vec{OA}$ and $\vec{OC}$ respectively.

- (a) Express side vectors $\vec{AB}$ and $\vec{CB}$ in terms of $\vec{OA}$ and $\vec{OC}$.
- (b) Express the position vector $\vec{OB}$ of vertex B in terms of $\vec{OA}$ and $\vec{OC}$.
- (c) Draw diagonal vector $\vec{AC}$ on the figure and express $\vec{AC}$ in terms of $\vec{OA}$ and $\vec{OC}$.
- (d) Draw diagonal vector $\vec{CA}$ on the figure. Describe the geometric relationship between $\vec{AC}$ and $\vec{CA}$.

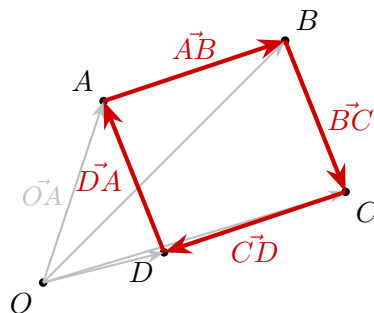


4. A particle starts at point A and moves along a quadrilateral path $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$, as shown below. The position vectors of points A, B, C , and D relative to an origin O are $\vec{OA}, \vec{OB}, \vec{OC}$, and $\vec{OD}$ respectively.

- (a) Express each displacement vector $\vec{AB}, \vec{BC}, \vec{CD}$, and $\vec{DA}$ in terms of $\vec{OA}, \vec{OB}, \vec{OC}$, and $\vec{OD}$.
- (b) Evaluate the vector sum:

$$\vec{AB} + \vec{BC} + \vec{CD} + \vec{DA}$$

- (c) Interpret the result in (b) geometrically in terms of net position change.



3.2 Arithmetic on Vectors

To distinguish vector directions from scalar coordinates, the standard convention in HKDSE M2 avoids using raw coordinate labels x and y as direction vectors. Instead, x and y remain scalar variables, and unit vectors $\hat{i}$ and $\hat{j}$ represent unit displacements along the positive x -axis and y -axis respectively.

Definition 3.2.1: Algebraic Definition of Planar Vector

Any 2D vector $\vec{v}$ can be uniquely expressed as a linear combination of the standard unit vectors $\hat{i}$ and $\hat{j}$:

$$\vec{v} = x\hat{i} + y\hat{j}$$

where $x, y \in \mathbb{R}$ are the scalar components of $\vec{v}$ along the x -axis and y -axis respectively.

Remark

The unit vectors $\hat{i}$ and $\hat{j}$ are linearly independent and orthogonal by definition.

Using this basis, componentwise arithmetic operations are defined straightforwardly.

Definition 3.2.2: Addition and Subtraction of Vectors

Let $\vec{u} = x_1\hat{i} + y_1\hat{j}$ and $\vec{v} = x_2\hat{i} + y_2\hat{j}$ be two vectors.

- The **sum** of $\vec{u}$ and $\vec{v}$ is defined componentwise as:

$$\vec{u} + \vec{v} = (x_1 + x_2)\hat{i} + (y_1 + y_2)\hat{j}$$

- The **difference** of $\vec{u}$ and $\vec{v}$ is defined componentwise as:

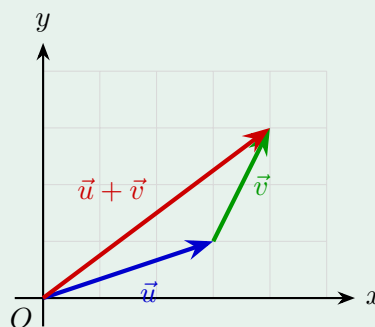
$$\vec{u} - \vec{v} = (x_1 - x_2)\hat{i} + (y_1 - y_2)\hat{j}$$

Example 3.2.1: Addition of Two Vectors

Let $\vec{u} = 3\hat{i} + \hat{j}$ and $\vec{v} = \hat{i} + 2\hat{j}$. Calculate $\vec{u} + \vec{v}$ algebraically.

Adding corresponding components:

$$\vec{u} + \vec{v} = (3 + 1)\hat{i} + (1 + 2)\hat{j} = 4\hat{i} + 3\hat{j}$$

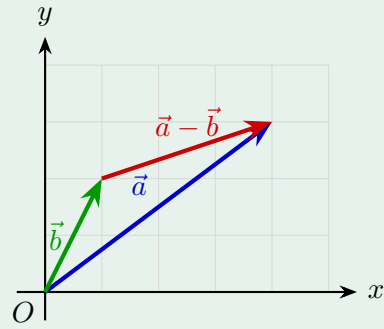


Example 3.2.2: Subtraction of Two Vectors

Let $\vec{a} = 4\hat{i} + 3\hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j}$. Calculate $\vec{a} - \vec{b}$ algebraically.

Subtracting corresponding components:

$$\vec{a} - \vec{b} = (4 - 1)\hat{i} + (3 - 2)\hat{j} = 3\hat{i} + \hat{j}$$


Exercise 3.2.1: Addition of Two Vectors

Given $\vec{p} = 2\hat{i} - 5\hat{j}$ and $\vec{q} = -3\hat{i} + 2\hat{j}$.

- Find $\vec{p} + \vec{q}$ in terms of $\hat{i}$ and $\hat{j}$.
- Calculate the magnitude $|\vec{p} + \vec{q}|$.

Exercise 3.2.2: Subtraction of Two Vectors

Given $\vec{u} = 5\hat{i} - 2\hat{j}$ and $\vec{v} = 2\hat{i} + 3\hat{j}$.

- Express $\vec{u} - \vec{v}$ and $\vec{v} - \vec{u}$ in terms of $\hat{i}$ and $\hat{j}$.
- Comment on the relationship between $\vec{u} - \vec{v}$ and $\vec{v} - \vec{u}$.

Subtracting a vector is geometrically equivalent to adding its reverse. Negation satisfies our intuition that negative signs represent reversed spatial orientation, operating seamlessly in $\mathbb{R}^2$.

Definition 3.2.3: Negation of a Vector

For any vector $\vec{v} = x\hat{i} + y\hat{j}$, its **negation** (or additive inverse) is denoted by $-\vec{v}$ and is defined as:

$$-\vec{v} = -x\hat{i} - y\hat{j}$$

Geometrically, $-\vec{v}$ has the same magnitude as $\vec{v}$ but opposite direction. For a directed line segment from A to B :

$$\vec{BA} = -\vec{AB}$$

Example 3.2.3: Displacement and Reverse Direction

If point A has position vector $\vec{OA} = 2\hat{i} + 3\hat{j}$ and point B has position vector $\vec{OB} = 5\hat{i} + 7\hat{j}$, then:

$$\vec{AB} = \vec{OB} - \vec{OA} = (5 - 2)\hat{i} + (7 - 3)\hat{j} = 3\hat{i} + 4\hat{j}$$

The reverse displacement vector from B to A is:

$$\vec{BA} = -\vec{AB} = -(3\hat{i} + 4\hat{j}) = -3\hat{i} - 4\hat{j}$$

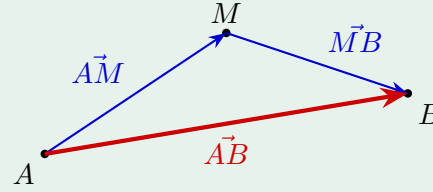
Example 3.2.4: Reversing a Vector Path

Consider moving along a path from point A through an intermediate point M to point B .

The net displacement is $\vec{AB} = \vec{AM} + \vec{MB}$.

Reversing the path gives:

$$\vec{BA} = \vec{BM} + \vec{MA} = -\vec{MB} - \vec{AM} = -\vec{AB}$$


Exercise 3.2.3: Path Vector Identification

In $\triangle ABC$, let $\vec{AB} = \hat{i} + 3\hat{j}$ and $\vec{BC} = 3\hat{i} - \hat{j}$.

(a) Express $\vec{AC}$ in terms of $\hat{i}$ and $\hat{j}$.

(b) Express $\vec{CA}$ and $\vec{CB}$ in terms of $\hat{i}$ and $\hat{j}$.

The key strength of vector algebra over standard coordinates lies in decentralization. Below are foundational theorems inheriting coordinate-free flexibility.

Theorem 3.2.1: Parallelogram Law of Vector Addition

If $ABCD$ is a parallelogram, then the diagonal vector $\vec{AC}$ is the sum of two adjacent side vectors sharing the same initial point:

$$\vec{AC} = \vec{AB} + \vec{AD}$$

In terms of position vectors relative to an arbitrary origin O :

$$\vec{OA} + \vec{OC} = \vec{OB} + \vec{OD}$$

Proof. Since $ABCD$ is a parallelogram, the opposite directed segments are equal in magnitude and parallel ($\vec{DC} = \vec{AB}$). By triangle addition:

$$\vec{AB} + \vec{AD} = \vec{AD} + \vec{DC} = \vec{AC}$$

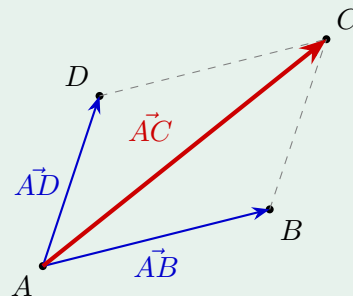
Expressing in position vectors gives $(\vec{OB} - \vec{OA}) + (\vec{OD} - \vec{OA}) = \vec{OC} - \vec{OA}$, which simplifies to $\vec{OA} + \vec{OC} = \vec{OB} + \vec{OD}$. $\square$

Example 3.2.5: Application of Parallelogram Law

$ABCD$ is a parallelogram with $\vec{AB} = 4\hat{i} + \hat{j}$ and $\vec{AD} = \hat{i} + 3\hat{j}$.

Using the Parallelogram Law:

$$\vec{AC} = \vec{AB} + \vec{AD} = (4\hat{i} + \hat{j}) + (\hat{i} + 3\hat{j}) = 5\hat{i} + 4\hat{j}$$



Exercise 3.2.4: Application of Parallelogram Law

In a parallelogram $OPQR$, O is the origin. Given $\vec{OP} = 3\hat{i} + 2\hat{j}$ and $\vec{OR} = -\hat{i} + 4\hat{j}$.

- Find the position vector $\vec{OQ}$ of vertex Q .
- Express the diagonal vector $\vec{PR}$ in terms of $\hat{i}$ and $\hat{j}$.

Theorem 3.2.2: Closed Polygon Vector Sum Theorem

For any closed sequence of points $P_1, P_2, P_3, \dots, P_n$ forming a polygon, the sum of directed vectors along the perimeter is equal to the zero vector $\vec{0}$:

$$\vec{P_1P_2} + \vec{P_2P_3} + \dots + \vec{P_nP_1} = \vec{0}$$

Proof. Applying the triangle law iteratively:

$$\vec{P_1P_2} + \vec{P_2P_3} + \dots + \vec{P_{n-1}P_n} + \vec{P_nP_1} = \vec{P_1P_n} + \vec{P_nP_1} = \vec{P_1P_1} = \vec{0}$$

□

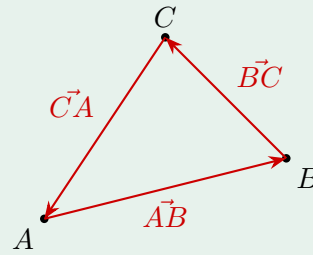
Example 3.2.6: Cyclic Path on Triangle

Show that for any triangle ABC , $\vec{AB} + \vec{BC} + \vec{CA} = \vec{0}$.

By triangle addition, $\vec{AB} + \vec{BC} = \vec{AC}$.

Thus:

$$\vec{AB} + \vec{BC} + \vec{CA} = \vec{AC} + \vec{CA} = \vec{AC} - \vec{AC} = \vec{0}$$


Exercise 3.2.5: Closed Polygon Sum

Given a pentagon $ABCDE$.

- Simplify $\vec{AB} + \vec{BC} + \vec{CD} + \vec{DE} + \vec{EA}$.
- Simplify $\vec{AC} + \vec{CE} + \vec{EA}$.

From the cyclic path theorem, we can further deduce vector representations for the **sum of roots of unity** and the **center of mass**.

Theorem 3.2.3: Sum of Position Vectors of Regular Polygon Vertices

Let $P_0, P_1, \dots, P_{n-1}$ be the vertices of a regular n -gon centered at origin O . The sum of their position vectors satisfies:

$$\sum_{k=0}^{n-1} \vec{OP_k} = \vec{0}$$

Proof. Translating each vector $\vec{OP_k}$ head-to-tail forms a closed regular n -gon due to rotational symmetry and equal magnitudes $|\vec{OP_k}|$. By Theorem 2.2.2, the sum around the closed loop vanishes, giving $\sum_{k=0}^{n-1} \vec{OP_k} = \vec{0}$. □

Theorem 3.2.4: Centroid of a System of Points

For a system of n points $A_1, A_2, \dots, A_n$ on a plane, the centroid G is the unique point satisfying:

$$\sum_{i=1}^n \vec{GA}_i = \vec{0}$$

Relative to an arbitrary origin O , the position vector of centroid G is given by:

$$\vec{OG} = \frac{1}{n} \sum_{i=1}^n \vec{OA}_i$$

Proof. Expanding position vectors relative to O :

$$\frac{1}{n} \sum_{i=1}^n \vec{OA}_i = \frac{1}{n} \sum_{i=1}^n (\vec{OG} + \vec{GA}_i) = \frac{1}{n} (n\vec{OG}) + \frac{1}{n} \sum_{i=1}^n \vec{GA}_i = \vec{OG} + \vec{0} = \vec{OG}$$

□

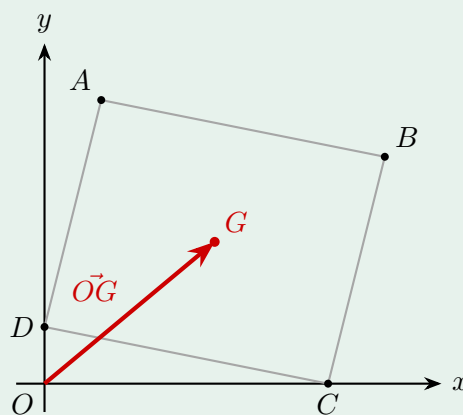
Example 3.2.7: Centroid of a Quadrilateral

Quadrilateral $ABCD$ has vertices with position vectors:

$$\vec{OA} = \hat{i} + 5\hat{j}, \quad \vec{OB} = 6\hat{i} + 4\hat{j},$$

$$\vec{OC} = 5\hat{i}, \quad \vec{OD} = \hat{j}$$

- Find the position vector $\vec{OG}$ of centroid G .
- Verify that $\vec{GA} + \vec{GB} + \vec{GC} + \vec{GD} = \vec{0}$.



Solution:

- By the centroid formula for $n = 4$:

$$\vec{OG} = \frac{1}{4}(\vec{OA} + \vec{OB} + \vec{OC} + \vec{OD}) = \frac{1}{4}[(\hat{i} + 5\hat{j}) + (6\hat{i} + 4\hat{j}) + 5\hat{i} + \hat{j}] = 3\hat{i} + \frac{5}{2}\hat{j}$$

- Computing displacement vectors from G :

$$\vec{GA} = -2\hat{i} + \frac{5}{2}\hat{j}, \quad \vec{GB} = 3\hat{i} + \frac{3}{2}\hat{j}, \quad \vec{GC} = 2\hat{i} - \frac{5}{2}\hat{j}, \quad \vec{GD} = -3\hat{i} - \frac{3}{2}\hat{j}$$

Summing gives $\vec{GA} + \vec{GB} + \vec{GC} + \vec{GD} = 0\hat{i} + 0\hat{j} = \vec{0}$.

Exercise 3.2.6: Centroid of a Triangle

The position vectors of the vertices of $\triangle ABC$ are $\vec{OA} = 3\hat{i} + 5\hat{j}$, $\vec{OB} = -\hat{i} + 2\hat{j}$, and $\vec{OC} = 4\hat{i} - \hat{j}$.

- Find the position vector $\vec{OG}$ of centroid G of $\triangle ABC$.
- Express displacement vectors $\vec{GA}$, $\vec{GB}$, and $\vec{GC}$ in component form.
- Verify algebraically that $\vec{GA} + \vec{GB} + \vec{GC} = \vec{0}$.

Follow-up Exercises**Level 1 (Basic)**

- Given vectors $\vec{u} = 4\hat{i} - 3\hat{j}$ and $\vec{v} = -2\hat{i} + 5\hat{j}$.
 - Find $\vec{u} + \vec{v}$ and $\vec{u} - \vec{v}$ in component form.
 - Calculate $|\vec{u} + \vec{v}|$.
 - Find the position vector of point A if $\vec{OA} = 2\vec{u} + \vec{v}$.
- In $\triangle PQR$, let M be the midpoint of side QR .
 - Express $\vec{PQ} + \vec{PR}$ in terms of median vector $\vec{PM}$.
 - Show that $\vec{PQ} + \vec{QR} + \vec{RP} = \vec{0}$.

Level 2 (Advanced)

- Let $ABCD$ be a quadrilateral, and let E and F be the midpoints of diagonals AC and BD respectively.
 - Express $\vec{AB} + \vec{AD}$ using position vectors relative to origin O .
 - Express $\vec{CB} + \vec{CD}$ using position vectors relative to origin O .
 - Prove that $\vec{AB} + \vec{AD} + \vec{CB} + \vec{CD} = 4\vec{EF}$.
- Let G be the centroid of $\triangle ABC$, and O be an arbitrary origin.
 - Given $\vec{OA} = 2\hat{i} + \hat{j}$, $\vec{OB} = 5\hat{i} + 4\hat{j}$, and $\vec{OG} = 3\hat{i} + 2\hat{j}$, find position vector $\vec{OC}$.
 - If P is a point such that $\vec{PA} + \vec{PB} + \vec{PC} = 3\hat{i} - 6\hat{j}$, express $\vec{PG}$ in component form.

3.3 Scaling Vectors

Since vectors represent spatial changes in position (displacements), we naturally encounter scenarios involving *motion in the same direction over different distances*.

Definition 3.3.1: Scalar Multiplication

For any vector $\vec{v} = x\hat{i} + y\hat{j}$ and any real scalar $k \in \mathbb{R}$, the **scalar multiplication** $k\vec{v}$ is defined componentwise as:

$$k\vec{v} = k(x\hat{i} + y\hat{j}) = (kx)\hat{i} + (ky)\hat{j}$$

Geometrically, $k\vec{v}$ rescales the magnitude of $\vec{v}$ by factor $|k|$:

- If $k > 0$, $k\vec{v}$ maintains the **same direction** as $\vec{v}$.
- If $k < 0$, $k\vec{v}$ points in the **opposite direction** of $\vec{v}$.
- If $k = 0$, $0\vec{v} = \vec{0}$ (the zero vector).

Remark

Scaling represents enlargement, contraction, or direction reversal. A scalar factor $k > 1$ represents positive expansion, $0 < k < 1$ represents positive contraction, and $k < 0$ reverses orientation.

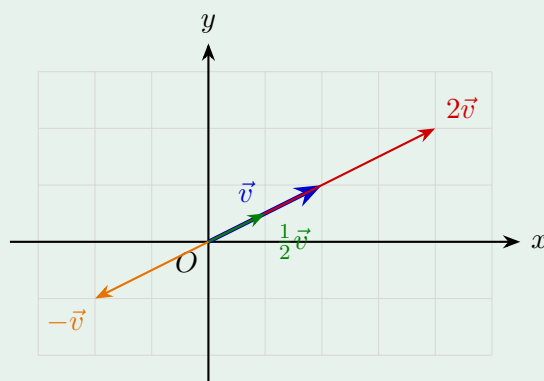
Visualizing scalar multiplication provides clear geometric intuition.

Example 3.3.1: Visualization of Scalar Multiplication

Let $\vec{v} = 2\hat{i} + \hat{j}$. Express $2\vec{v}$, $\frac{1}{2}\vec{v}$, and $-\vec{v}$ in component form.

Algebraically:

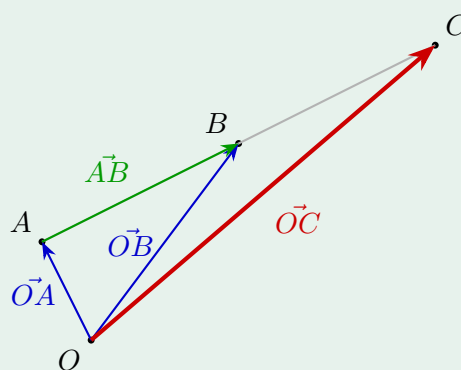
$$2\vec{v} = 4\hat{i} + 2\hat{j}, \quad \frac{1}{2}\vec{v} = \hat{i} + 0.5\hat{j}, \quad -\vec{v} = -2\hat{i} - \hat{j}$$



Example 3.3.2: Geometric Application of Scalar Multiplication

Let $\vec{OA} = \hat{i} + 2\hat{j}$ and $\vec{OB} = 4\hat{i} + 8\hat{j}$. Point C lies on line segment AB extended such that $\vec{AC} = 3\vec{AB}$. Find position vector $\vec{OC}$.

First, $\vec{AB} = \vec{OB} - \vec{OA} = 3\hat{i} + 6\hat{j}$. Then $\vec{AC} = 3(3\hat{i} + 6\hat{j}) = 9\hat{i} + 18\hat{j}$. Thus, $\vec{OC} = \vec{OA} + \vec{AC} = 10\hat{i} + 20\hat{j}$.



Exercise 3.3.1: Application of Scalar Multiplication

Given points $P(2, -1)$ and $Q(8, 11)$ with respect to origin O .

- Find $\vec{PQ}$ in terms of $\hat{i}$ and $\hat{j}$.
- Point R lies on line segment PQ such that $\vec{PR} = \frac{1}{3}\vec{PQ}$. Find $\vec{OR}$ and coordinates of R .
- Determine whether $\vec{OP}$ and $\vec{OR}$ are parallel.

Having defined scaling, we evaluate vector length using Pythagoras' Theorem.

Theorem 3.3.1: Magnitude of a Vector

For any vector $\vec{v} = x\hat{i} + y\hat{j}$, the **magnitude** (length) of $\vec{v}$, denoted by $|\vec{v}|$, is:

$$|\vec{v}| = \sqrt{x^2 + y^2}$$

For any scalar $k \in \mathbb{R}$, $|k\vec{v}| = |k||\vec{v}|$.

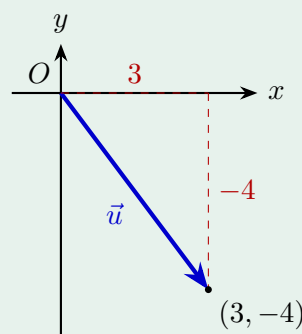
Example 3.3.3: Magnitude Calculation

Calculate the magnitude of $\vec{u} = 3\hat{i} - 4\hat{j}$ and $\vec{v} = -5\hat{i} + 12\hat{j}$.

Applying the distance formula:

$$|\vec{u}| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$$

$$|\vec{v}| = \sqrt{(-5)^2 + 12^2} = \sqrt{169} = 13$$


Exercise 3.3.2: Magnitude Equation

Given vector $\vec{a} = (k - 1)\hat{i} + 4\hat{j}$, where k is a real constant.

- Express $|\vec{a}|$ in terms of k .
- If $|\vec{a}| = 5$, find all possible values of k .

Standardizing vectors leads to the notion of a **unit vector** along any given direction.

Definition 3.3.2: Unit Vector

A vector of magnitude 1 is called a **unit vector**. A unit vector in the direction of a non-zero vector $\vec{v}$ is denoted as $\hat{v}$, satisfying $|\hat{v}| = 1$.

Remark

The basis vectors $\hat{i}$ and $\hat{j}$ are primary unit vectors along the positive x -axis and y -axis.

Theorem 3.3.2: Standardization (Normalization) of a Vector

For any non-zero vector $\vec{v}$, the unit vector $\hat{v}$ pointing in the direction of $\vec{v}$ is:

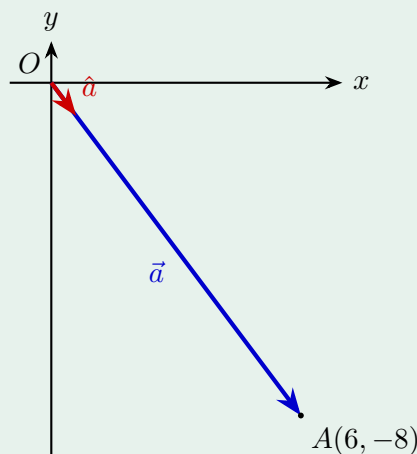
$$\hat{v} = \frac{\vec{v}}{|\vec{v}|} = \frac{1}{|\vec{v}|}\vec{v}$$

Example 3.3.4: Standardization Example

Find unit vector $\hat{a}$ in the direction of $\vec{a} = 6\hat{i} - 8\hat{j}$.

First, $|\vec{a}| = \sqrt{6^2 + (-8)^2} = 10$. Dividing $\vec{a}$ by its magnitude gives:

$$\hat{a} = \frac{6\hat{i} - 8\hat{j}}{10} = \frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}$$


Exercise 3.3.3: Unit Vector Determination

Given $\vec{w} = -3\hat{i} + 4\hat{j}$.

- Find the unit vector in the direction of $\vec{w}$.
- Find the unit vector pointing in the opposite direction of $\vec{w}$.

Comparing vector magnitudes allows us to express proportional relationships.

Example 3.3.5: Parallel Vector Rescaling

Given $\vec{a} = 3\hat{i} + 4\hat{j}$ and $\vec{b} = 6\hat{i} + 8\hat{j}$. Show that $\vec{b}$ is parallel to $\vec{a}$ and determine ratio $|\vec{b}| : |\vec{a}|$.

Solution: Factoring out 2 from $\vec{b}$:

$$\vec{b} = 6\hat{i} + 8\hat{j} = 2(3\hat{i} + 4\hat{j}) = 2\vec{a}$$

Since $\vec{b} = k\vec{a}$ with scalar $k = 2 > 0$, $\vec{b}$ is parallel to $\vec{a}$ in the same direction. Evaluating magnitudes: $|\vec{a}| = 5$ and $|\vec{b}| = 10$. Thus, $|\vec{b}| : |\vec{a}| = 2 : 1$.

Example 3.3.6: Constructing a Vector with Specified Direction and Magnitude

Find a vector $\vec{u}$ in the direction of $\vec{v} = 5\hat{i} - 12\hat{j}$ with magnitude equal to $|\vec{w}|$, where $\vec{w} = 3\hat{i} + 4\hat{j}$.

Solution: 1. Required magnitude: $|\vec{w}| = \sqrt{3^2 + 4^2} = 5$. 2. Direction unit vector: $|\vec{v}| = 13 \implies \hat{v} = \frac{5}{13}\hat{i} - \frac{12}{13}\hat{j}$. 3. Desired vector: $\vec{u} = |\vec{w}|\hat{v} = 5 \left(\frac{5}{13}\hat{i} - \frac{12}{13}\hat{j} \right) = \frac{25}{13}\hat{i} - \frac{60}{13}\hat{j}$.

Exercise 3.3.4: Rescaling Vectors

Given $\vec{p} = 8\hat{i} - 6\hat{j}$.

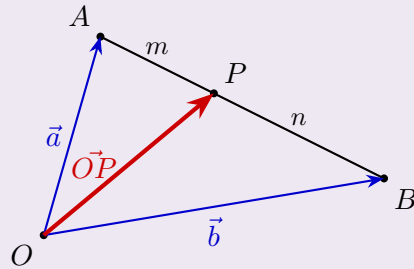
- (a) Find a vector $\vec{q}$ in the direction of $\vec{p}$ with magnitude 15.
- (b) Find a vector $\vec{r}$ in the opposite direction of $\vec{p}$ with magnitude 20.

We conclude this section by reformulating the standard coordinate section formula in vector terms.

Theorem 3.3.3: Section Formula for Internal Division

Let A and B have position vectors $\vec{OA} = \vec{a}$ and $\vec{OB} = \vec{b}$ relative to origin O . If point P divides line segment AB internally in ratio $m : n$ (i.e. $AP : PB = m : n$), then:

$$\vec{OP} = \frac{n\vec{OA} + m\vec{OB}}{m + n} = \frac{n\vec{a} + m\vec{b}}{m + n}$$



Proof. Since $AP : PB = m : n$, we have $\vec{AP} = \frac{m}{m+n}\vec{AB}$. Expressing in position vectors:

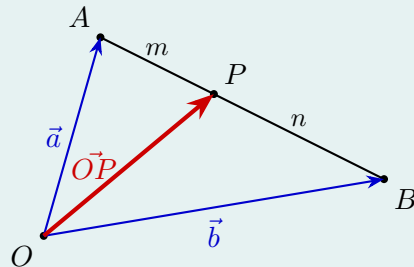
$$\vec{OP} - \vec{OA} = \frac{m}{m+n}(\vec{OB} - \vec{OA}) \implies \vec{OP} = \left(1 - \frac{m}{m+n}\right)\vec{OA} + \frac{m}{m+n}\vec{OB} = \frac{n\vec{a} + m\vec{b}}{m+n}$$

□

Proposition 3.3.1: Section Formula (General Origin Independence)

The section formula holds independently of choice of origin O :

$$\vec{OP} = \frac{n\vec{a} + m\vec{b}}{m + n}$$

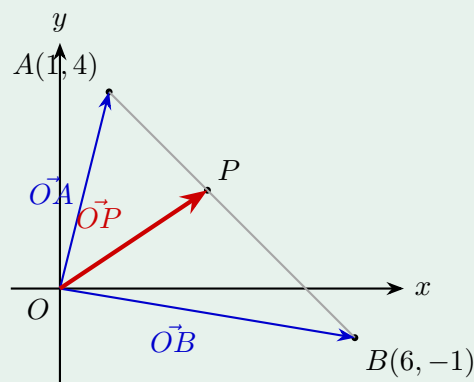


Proof. Follows directly from Theorem 3.3.3 by vector addition $\vec{OP} = \vec{OA} + \vec{AP}$.

□

Example 3.3.7: Section Formula Calculation

Position vectors of A and B relative to O are $\vec{OA} = \hat{i} + 4\hat{j}$ and $\vec{OB} = 6\hat{i} - \hat{j}$. Point P lies on AB such that $AP : PB = 2 : 3$. Find coordinates of P .



Solution: Applying the section formula with $m = 2$ and $n = 3$:

$$\vec{OP} = \frac{3\vec{OA} + 2\vec{OB}}{2 + 3} = \frac{3(\hat{i} + 4\hat{j}) + 2(6\hat{i} - \hat{j})}{5} = \frac{15\hat{i} + 10\hat{j}}{5} = 3\hat{i} + 2\hat{j}$$

Thus, coordinates of P are $(3, 2)$.

Exercise 3.3.5: Section Formula Application

Coordinates of points are $A(-2, 7)$ and $B(8, -3)$.

- Find position vector $\vec{OP}$ of point P dividing AB internally in ratio $3 : 2$.
- If M is the midpoint of AP , find position vector $\vec{OM}$ and coordinates of M .

Follow-up Practice**Level 1 (Basic)**

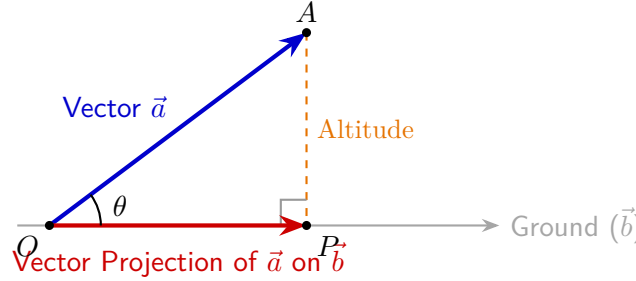
1. Given vectors $\vec{u} = 6\hat{i} - 8\hat{j}$ and $\vec{v} = -2\hat{i} + \hat{j}$.
 - (a) Find $3\vec{u} - 2\vec{v}$ in component form.
 - (b) Calculate $|\vec{u}|$.
 - (c) Find unit vector $\hat{u}$ in the direction of $\vec{u}$.
2. Given points $A(1, 2)$, $B(4, 8)$, and $C(6, 12)$.
 - (a) Find displacement vectors $\vec{AB}$ and $\vec{BC}$ in component form.
 - (b) Show that $\vec{BC} = k\vec{AB}$ for some scalar k , and deduce that A, B , and C are collinear.

Level 2 (Advanced)

3. Position vectors are $\vec{OP} = 4\hat{i} + 3\hat{j}$ and $\vec{OQ} = (t + 1)\hat{i} + 12\hat{j}$, where $t \in \mathbb{R}$.
 - (a) If $\vec{OQ}$ is parallel to $\vec{OP}$, find value of t .
 - (b) Using t from (a), find vector $\vec{w}$ pointing in opposite direction of $\vec{PQ}$ with magnitude 10.
4. In $\triangle ABC$, point D lies on side BC such that $BD : DC = 2 : 1$. Let $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$, and $\vec{OC} = \vec{c}$.
 - (a) Express $\vec{BC}$ in terms of $\vec{b}$ and $\vec{c}$.
 - (b) Prove that $\vec{OD} = \frac{\vec{b} + 2\vec{c}}{3}$.
 - (c) If E is the midpoint of AD , express $\vec{OE}$ in terms of $\vec{a}$, $\vec{b}$, and $\vec{c}$.

3.4 Scalar Product (Dot Product)

In vector geometry, projections allow us to determine the component of a vector along a specific direction, analogous to shadows cast onto a ground plane.



Formalizing vector projection analytically:

Proposition 3.4.1: Vector Projection

Let $\vec{a}$ and $\vec{b}$ be vectors with included angle θ . The **vector projection** of $\vec{a}$ onto non-zero vector $\vec{b}$ is:

$$\vec{p} = \left(\frac{|\vec{a}| \cos \theta}{|\vec{b}|} \right) \vec{b}$$

Proof. The scalar projection (signed length) of $\vec{a}$ along $\vec{b}$ is $|\vec{a}| \cos \theta$. Multiplying by unit vector $\frac{\vec{b}}{|\vec{b}|}$ yields $\vec{p} = \left(\frac{|\vec{a}| \cos \theta}{|\vec{b}|} \right) \vec{b}$. $\square$

Since calculating θ explicitly can yield irrational angles, we reformulate vector projection into a coordinate-friendly product. Notice that when $\theta = 0$, projection reduces to self-projection ($\vec{p} = \vec{b}$), whereas when vectors are orthogonal ($\theta = \pi/2$), the projection vanishes ($\vec{p} = \vec{0}$).

Multiplying the projected length by the magnitude of the ground vector yields the scalar value $|\vec{a}| |\vec{b}| \cos \theta$, giving our fundamental definition:

Definition 3.4.1: Scalar Product (Dot Product)

Let $\vec{a}$ and $\vec{b}$ be two vectors with included angle θ ($0 \leq \theta \leq \pi$). The **scalar product** (or **dot product**) $\vec{a} \cdot \vec{b}$ is defined as:

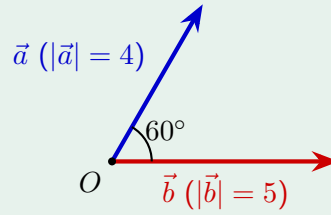
$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

Remark

Physicist Josiah Willard Gibbs introduced dot notation to represent a product yielding a 0-dimensional (scalar) quantity.

Example 3.4.1: Dot Product Calculation

Given $|\vec{a}| = 4$, $|\vec{b}| = 5$, and included angle $\theta = 60^\circ$. Calculate $\vec{a} \cdot \vec{b}$.



Solution:

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos 60^\circ = (4)(5) \left(\frac{1}{2}\right) = 10$$

Exercise 3.4.1: Angle between Vectors

Given $|\vec{u}| = 6$, $|\vec{v}| = 2\sqrt{3}$, and $\vec{u} \cdot \vec{v} = -18$.

- Find value of $\cos \theta$, where θ is the included angle.
- Determine angle θ in degrees.

Proposition 3.4.2: Properties of Scalar Product

For any vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$:

- Commutativity: $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$.
- Self Dot Product: $\vec{a} \cdot \vec{a} = |\vec{a}|^2$.
- Distributivity: $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$.
- Orthogonality: Non-zero vectors $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$.

Proof. (a) $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta = |\vec{b}||\vec{a}| \cos \theta = \vec{b} \cdot \vec{a}$.

(b) $\vec{a} \cdot \vec{a} = |\vec{a}||\vec{a}| \cos 0^\circ = |\vec{a}|^2$.

□

Caution

The scalar product is non-associative: $(\vec{a} \cdot \vec{b})\vec{c} \neq \vec{a}(\vec{b} \cdot \vec{c})$. The left side is parallel to $\vec{c}$, whereas the right side is parallel to $\vec{a}$.

Example 3.4.2: Algebraic Manipulation of Scalar Product

Let $|\vec{a}| = 3$, $|\vec{b}| = 4$, and $|\vec{a} + \vec{b}| = \sqrt{37}$.

- Find scalar product $\vec{a} \cdot \vec{b}$.
- Find angle θ between $\vec{a}$ and $\vec{b}$.

Solution:

(a) Expanding $|\vec{a} + \vec{b}|^2$:

$$|\vec{a} + \vec{b}|^2 = (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2$$

$$37 = 3^2 + 2\vec{a} \cdot \vec{b} + 4^2 \implies 37 = 25 + 2\vec{a} \cdot \vec{b} \implies \vec{a} \cdot \vec{b} = 6$$

(b) Using $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$:

$$6 = (3)(4) \cos \theta \implies \cos \theta = \frac{1}{2} \implies \theta = 60^\circ$$

Exercise 3.4.2: Algebraic Manipulation

Given $|\vec{p}| = 2$, $|\vec{q}| = 5$, and included angle 120° .

- Evaluate $\vec{p} \cdot \vec{q}$.
- Find magnitude $|\vec{p} - 2\vec{q}|$.

For orthogonal unit vectors $\hat{i}$ and $\hat{j}$:

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = 1 \quad \text{and} \quad \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{i} = 0$$

Proposition 3.4.3: Componentwise Scalar Product

Let $\vec{a} = a_x \hat{i} + a_y \hat{j}$ and $\vec{b} = b_x \hat{i} + b_y \hat{j}$. Then:

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y$$

Example 3.4.3: Componentwise Scalar Product Example

Given $\vec{a} = 2\hat{i} + 3\hat{j}$ and $\vec{b} = 4\hat{i} - \hat{j}$.

- Calculate $\vec{a} \cdot \vec{b}$.
- Find angle θ between $\vec{a}$ and $\vec{b}$ correct to 1 decimal place.

Solution:

$$(a) \quad \vec{a} \cdot \vec{b} = (2)(4) + (3)(-1) = 8 - 3 = 5.$$

$$(b) \quad |\vec{a}| = \sqrt{13}, |\vec{b}| = \sqrt{17}.$$

$$\cos \theta = \frac{5}{\sqrt{13}\sqrt{17}} = \frac{5}{\sqrt{221}} \approx 0.3363 \implies \theta \approx 70.3^\circ$$

Exercise 3.4.3: Componentwise Scalar Product

Given $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = 2\hat{i} - 4\hat{j}$.

- If $\vec{u} \perp \vec{v}$, find k .
- If $k = 1$, calculate $\vec{u} \cdot \vec{v}$ and $\cos \theta$.

Component algebra allows vector projection to be written without referencing angle θ explicitly.

Theorem 3.4.1: Projection Vector Formula

The vector projection $\vec{p}$ of vector $\vec{a}$ onto vector $\vec{b}$ is:

$$\vec{p} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b} = \left(\frac{\vec{a} \cdot \vec{b}}{\vec{b} \cdot \vec{b}} \right) \vec{b}$$

Proof. Substituting $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$ into Proposition 3.4.1 gives:

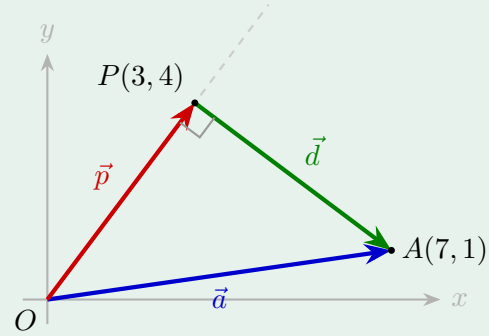
$$\vec{p} = \left(\frac{|\vec{a}|}{|\vec{b}|} \cdot \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} \right) \vec{b} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b}$$

□

Example 3.4.4: Vector Projection and Orthogonal Decomposition

Let $\vec{a} = 7\hat{i} + \hat{j}$ and $\vec{b} = 3\hat{i} + 4\hat{j}$.

- Find vector projection $\vec{p}$ of $\vec{a}$ onto $\vec{b}$.
- Find perpendicular component $\vec{d} = \vec{a} - \vec{p}$ and verify $\vec{d} \perp \vec{b}$.



Solution:

- $\vec{a} \cdot \vec{b} = 21 + 4 = 25$ and $|\vec{b}|^2 = 25$. Thus, $\vec{p} = \frac{25}{25}(3\hat{i} + 4\hat{j}) = 3\hat{i} + 4\hat{j}$.
- $\vec{d} = (7\hat{i} + \hat{j}) - (3\hat{i} + 4\hat{j}) = 4\hat{i} - 3\hat{j}$. Testing dot product: $\vec{d} \cdot \vec{b} = 12 - 12 = 0 \implies \vec{d} \perp \vec{b}$.

Exercise 3.4.4: 2D Projection Decomposition

Given $\vec{u} = 5\hat{i} + 10\hat{j}$ and $\vec{v} = 4\hat{i} - 3\hat{j}$.

- Find vector projection $\vec{p}$ of $\vec{u}$ onto $\vec{v}$.
- Decompose $\vec{u} = \vec{u}_1 + \vec{u}_2$, where $\vec{u}_1 // \vec{v}$ and $\vec{u}_2 \perp \vec{v}$.

Proposition 3.4.4: Cosine Formula via Scalar Product

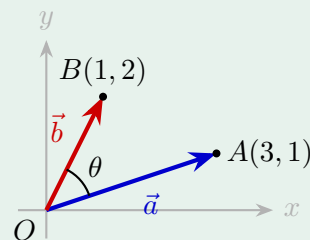
Let $\vec{a}$ and $\vec{b}$ be non-zero vectors with included angle θ . Then:

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$$

Proof. Direct rearrangement of Definition 3.4.1. □

Example 3.4.5: Angle Calculation Example

Find exact angle θ between $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j}$.



Solution: $\vec{a} \cdot \vec{b} = 5$, $|\vec{a}| = \sqrt{10}$, $|\vec{b}| = \sqrt{5}$.

$$\cos \theta = \frac{5}{\sqrt{10}\sqrt{5}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}} \implies \theta = 45^\circ$$

Exercise 3.4.5: Angle Determination

- (a) Find exact angle θ between $\vec{p} = \hat{i} + \sqrt{3}\hat{j}$ and $\vec{q} = \sqrt{3}\hat{i} + \hat{j}$.
 (b) Given $A(1, 2)$, $B(4, 5)$, and $C(1, 6)$, find $\vec{AB}$, $\vec{AC}$, and $\angle BAC$.

Follow-up Practice

Level 1 (Basic)

1. Given $\vec{a} = 2\hat{i} + 5\hat{j}$ and $\vec{b} = k\hat{i} - 4\hat{j}$.
 - (a) Express $\vec{a} \cdot \vec{b}$ in terms of k .
 - (b) If $\vec{a} \perp \vec{b}$, find k .
 - (c) If $k = 3$, calculate angle between $\vec{a}$ and $\vec{b}$ to the nearest degree.
2. Unit vectors $\hat{u}$ and $\hat{v}$ have included angle 60° .
 - (a) Evaluate $\hat{u} \cdot \hat{v}$.
 - (b) Show that $|\hat{u} + \hat{v}| = \sqrt{3}$.
 - (c) Evaluate $|\hat{u} - \hat{v}|$.

Level 2 (Advanced)

3. Given points $A(6, 8)$ and $B(8, 0)$ relative to origin O .
 - (a) Find position vectors $\vec{OA}$ and $\vec{OB}$.
 - (b) Find vector projection $\vec{p}$ of $\vec{OA}$ onto $\vec{OB}$.
 - (c) Find perpendicular distance from point A to line OB .
 - (d) Verify $(\vec{OA} - \vec{p}) \perp \vec{OB}$.
4. In $\triangle ABC$, let $\vec{c} = \vec{AB}$, $\vec{a} = \vec{BC}$, and $\vec{b} = \vec{CA}$.
 - (a) Using $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, prove Law of Cosines via scalar product:

$$|\vec{a}|^2 = |\vec{b}|^2 + |\vec{c}|^2 - 2|\vec{b}||\vec{c}|\cos A$$

- (b) Given $|\vec{b}| = 5$, $|\vec{c}| = 8$, and $\angle A = 60^\circ$, find $|\vec{a}|$.

3.5 Vector Product and Area in 2D

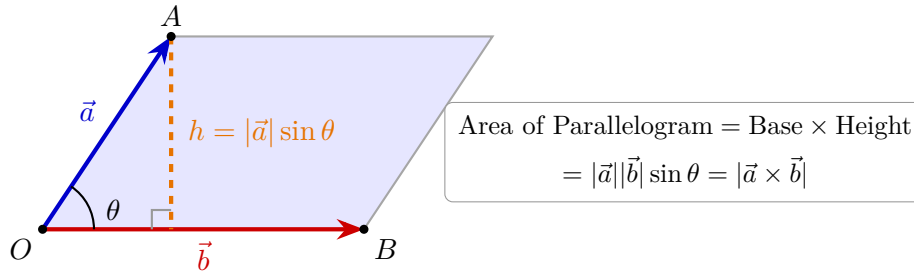
While scalar product measures parallel alignment, vector product (cross product) measures oriented area spanned by vectors across orthogonal directions.

Definition 3.5.1: 2D Vector Product / Area Product

For 2D vectors $\vec{a}$ and $\vec{b}$ in the xy -plane, the **area product** (or 2D cross product) $\vec{a} \times \vec{b}$ represents a directed area element orthogonal to the plane:

$$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin \theta (\hat{i} \times \hat{j})$$

where θ is the directed angle from $\vec{a}$ to $\vec{b}$.



Evaluating componentwise, parallel cross products vanish: $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \vec{0}$.

Proposition 3.5.1: Component Expansion of Area Product

Let $\vec{a} = a_x \hat{i} + a_y \hat{j}$ and $\vec{b} = b_x \hat{i} + b_y \hat{j}$. Then:

$$\vec{a} \times \vec{b} = a_x b_y (\hat{i} \times \hat{j}) + a_y b_x (\hat{j} \times \hat{i})$$

Proof. Expanding distributively:

$$\vec{a} \times \vec{b} = (a_x \hat{i} + a_y \hat{j}) \times (b_x \hat{i} + b_y \hat{j}) = a_x b_x (\hat{i} \times \hat{i}) + a_x b_y (\hat{i} \times \hat{j}) + a_y b_x (\hat{j} \times \hat{i}) + a_y b_y (\hat{j} \times \hat{j})$$

Since $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \vec{0}$, the result follows. $\square$

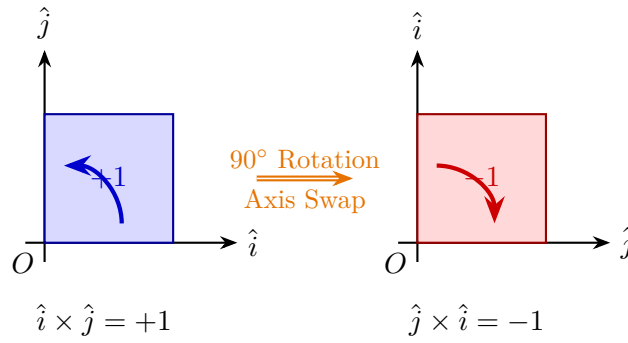
If we see $\hat{i} \times \hat{j}$ as the area of the square with sides $\hat{i}$ and $\hat{j}$, then $\hat{j} \times \hat{i}$ is also the area of the square with sides $\hat{i}$ and $\hat{j}$, but with different **orientation**. That is, the order of multiplication affects the calculation of area.

Remark (The calculus viewpoint of area product)

We can define $\vec{a} \times \vec{b} := \int \vec{a} d\vec{b}$. Although a formal proof is beyond our scope, it captures the underlying idea of accumulating area.

In fact, this makes sense: if one integrates area by value of $\hat{i}$ along $\hat{j}$, that is, $\int \hat{i} d\hat{j}$, the mechanism should be identical to $\int \hat{j} d(-\hat{i}) = -\int \hat{j} d\hat{i}$. This yields:

$$\hat{i} \times \hat{j} = -\hat{j} \times \hat{i}$$


Remark

This property is called **anti-commutativity**.

Theorem 3.5.1: Component Formula for 2D Cross Product

Let $\vec{a} = a_x \hat{i} + a_y \hat{j}$ and $\vec{b} = b_x \hat{i} + b_y \hat{j}$. Then:

$$\vec{a} \times \vec{b} = (a_x b_y - a_y b_x)(\hat{i} \times \hat{j})$$

Proof. Substituting anti-commutativity property $\hat{j} \times \hat{i} = -(\hat{i} \times \hat{j})$ into Proposition 3.5.1 yields:

$$\vec{a} \times \vec{b} = a_x b_y (\hat{i} \times \hat{j}) - a_y b_x (\hat{i} \times \hat{j}) = (a_x b_y - a_y b_x)(\hat{i} \times \hat{j})$$

□

Proposition 3.5.2: Anti-commutativity of Area Product

For any vectors $\vec{a}$ and $\vec{b}$:

$$\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$$

Proof. Using component formula: $\vec{b} \times \vec{a} = (b_x a_y - b_y a_x)(\hat{i} \times \hat{j}) = -(a_x b_y - a_y b_x)(\hat{i} \times \hat{j}) = -\vec{a} \times \vec{b}$. □

Taking magnitude yields non-negative geometric area directly.

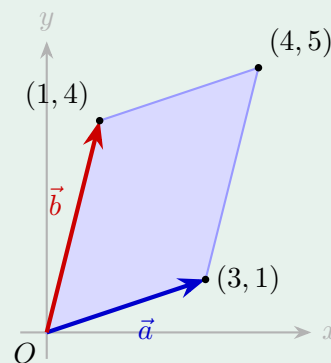
Example 3.5.1: Area of Parallelogram Spanned by Vectors

Adjacent side vectors of a parallelogram are $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 4\hat{j}$. Find area.

Using component formula:

$$\vec{a} \times \vec{b} = ((3)(4) - (1)(1))(\hat{i} \times \hat{j}) = 11(\hat{i} \times \hat{j})$$

$$\text{Area} = |\vec{a} \times \vec{b}| = 11 \text{ sq. units.}$$



Exercise 3.5.1: Area of Parallelogram

Given $\vec{u} = 5\hat{i} - 2\hat{j}$ and $\vec{v} = 2\hat{i} + 3\hat{j}$.

- Compute $\vec{u} \times \vec{v}$.
- Find area of parallelogram spanned by $\vec{u}$ and $\vec{v}$.

Example 3.5.2: Area of Triangle Given Vertices

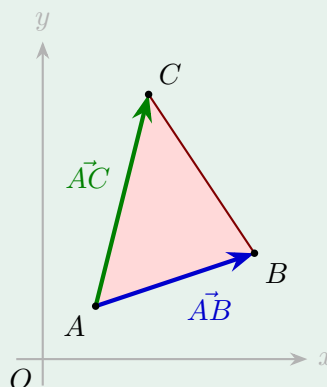
Vertices are $A(1, 1)$, $B(4, 2)$, and $C(2, 5)$.

Find area of $\triangle ABC$.

$\vec{AB} = 3\hat{i} + \hat{j}$ and $\vec{AC} = \hat{i} + 4\hat{j}$.

$$\vec{AB} \times \vec{AC} = (12 - 1)(\hat{i} \times \hat{j}) = 11(\hat{i} \times \hat{j})$$

$$\text{Area} = \frac{1}{2}|\vec{AB} \times \vec{AC}| = \frac{11}{2} = 5.5 \text{ sq. units}$$


Exercise 3.5.2: Area of Triangle

Given $P(-1, 2)$, $Q(3, -2)$, and $R(5, 4)$.

- Express $\vec{PQ}$ and $\vec{PR}$ in component form.
- Calculate $\vec{PQ} \times \vec{PR}$ and area of $\triangle PQR$.

Determinants provide an elegant matrix computational method for polygon areas.

Proposition 3.5.3: Determinant Formula for Triangle Area

Let $\vec{a} = a_x\hat{i} + a_y\hat{j}$ and $\vec{b} = b_x\hat{i} + b_y\hat{j}$. The area of $\triangle OAB$ is:

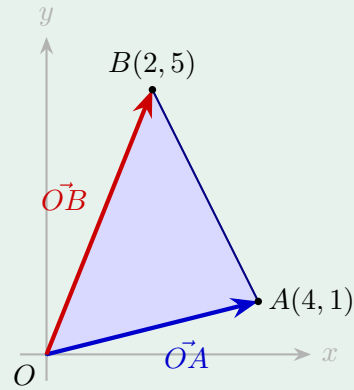
$$\text{Area}(\triangle OAB) = \frac{1}{2} \begin{vmatrix} a_x & a_y \\ b_x & b_y \end{vmatrix} = \frac{1}{2}|a_x b_y - a_y b_x|$$

Proof. The area of $\triangle OAB$ is half the magnitude of parallelogram vector area $|\vec{a} \times \vec{b}| = |a_x b_y - a_y b_x|$. Expressing as a 2×2 determinant yields the result. $\square$

Example 3.5.3: Determinant Area Example

Find area of $\triangle OAB$ with $A(4, 1)$ and $B(2, 5)$ relative to origin O .

$$\text{Area} = \frac{1}{2} \left\| \begin{vmatrix} 4 & 1 \\ 2 & 5 \end{vmatrix} \right\| = \frac{1}{2} |20 - 2| = 9 \text{ sq. units}$$


Exercise 3.5.3: Determinant Area Calculation

Given $A(-2, 6)$ and $B(5, 3)$ relative to origin O .

- (a) Find area of $\triangle OAB$.
- (b) Point $C(k, 4)$ lies in Quadrant I such that $\text{Area}(\triangle OAC) = 10$. Find k .

Proposition 3.5.4: Shoelace Formula for Polygon Area

Let $A_1(x_1, y_1), A_2(x_2, y_2), \dots, A_n(x_n, y_n)$ be vertices of an n -gon in counter-clockwise order. Area S is:

$$S = \frac{1}{2} \left| \sum_{t=1}^n (x_t y_{t+1} - x_{t+1} y_t) \right| = \frac{1}{2} \left\| \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_n & y_n \\ x_1 & y_1 \end{vmatrix} \right\|$$

where $(x_{n+1}, y_{n+1}) = (x_1, y_1)$.

Proof. Decomposing the polygon into n oriented triangular slices centered at origin O :

$$\text{Area} = \frac{1}{2} \left| \sum_{t=1}^n O\vec{A}_t \times O\vec{A}_{t+1} \right| = \frac{1}{2} \left| \sum_{t=1}^n (x_t y_{t+1} - x_{t+1} y_t) \right|$$

□

Example 3.5.4: Shoelace Formula Application

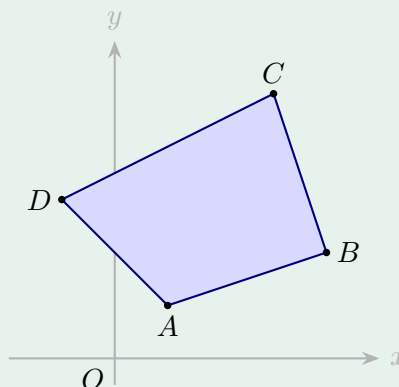
Find area of quadrilateral $ABCD$ with vertices $A(1, 1)$, $B(4, 2)$, $C(3, 5)$, and $D(-1, 3)$.

$$\text{Area} = \frac{1}{2} \begin{vmatrix} 1 & 1 \\ 4 & 2 \\ 3 & 5 \\ -1 & 3 \\ 1 & 1 \end{vmatrix}$$

Down-right sum: $(2) + (20) + (9) + (-1) = 30$.

Up-right sum: $(4) + (6) + (-5) + (3) = 8$.

$$\text{Area} = \frac{1}{2} |30 - 8| = 11 \text{ sq. units}$$

**Exercise 3.5.4: Polygon Area Calculation**

Given pentagon $ABCDE$ with vertices $A(0, 1)$, $B(3, 0)$, $C(5, 3)$, $D(2, 6)$, and $E(-2, 4)$.

- Calculate area using the Shoelace formula.
- Verify by decomposing into triangular slices from origin O .

Proposition 3.5.5: Sine Formula via Area Product

Let θ be the included angle between non-zero vectors $\vec{a}$ and $\vec{b}$. Then:

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$$

Proof. Follows directly from Definition 3.5.1 by taking magnitudes. □

Caution

Due to rotational sign sensitivity, cross products require careful attention to vector ordering when computing angles.

Follow-up Practice**Level 1 (Basic)**

- Given $\vec{a} = 3\hat{i} - 2\hat{j}$ and $\vec{b} = \hat{i} + 4\hat{j}$.
 - Calculate $\vec{a} \times \vec{b}$.
 - Find area of parallelogram formed by $\vec{a}$ and $\vec{b}$.
 - Find area of triangle formed by $\vec{a}$ and $\vec{b}$.
- Given $A(3, 1)$, $B(1, 4)$, and $C(-2, 2)$ relative to origin O .
 - Find area of $\triangle OAB$.
 - Find area of quadrilateral $OABC$ using Shoelace formula.

Level 2 (Advanced)

3. Points are $A(1, 2)$, $B(4, k)$, and $C(2, 5)$, where $k > 0$.
- (a) Express $\vec{AB}$ and $\vec{AC}$ in component form.
 - (b) Compute $\vec{AB} \times \vec{AC}$ in terms of k .
 - (c) If $\text{Area}(\triangle ABC) = 6$, find k .
 - (d) Find $\sin \angle BAC$.
4. Let $\triangle ABC$ have vertices $A(0, 0)$, $B(x_1, y_1)$, and $C(x_2, y_2)$.
- (a) Express $\vec{AB} \cdot \vec{AC}$ and $|\vec{AB} \times \vec{AC}|$ in components.
 - (b) Verify 2D Lagrange's Identity:

$$|\vec{AB}|^2 |\vec{AC}|^2 = (\vec{AB} \cdot \vec{AC})^2 + |\vec{AB} \times \vec{AC}|^2$$

- (c) Deduce $\sin^2 A + \cos^2 A = 1$.
- (d) If $B(4, 0)$ and $C(2, 2\sqrt{3})$, evaluate area of $\triangle ABC$ and $\sin A$.

3.6 Comprehensive Exercises

- Given non-zero non-parallel vectors $\mathbf{a}$ and $\mathbf{b}$. Points A, B, C , and D have position vectors relative to origin O given by $\vec{OA} = \mathbf{a} + 2\mathbf{b}$, $\vec{OB} = 3\mathbf{a} - k\mathbf{b}$, $\vec{OC} = 2\mathbf{a} + \mathbf{b}$, and $\vec{OD} = (k+1)\mathbf{a} + 5\mathbf{b}$, where k is a scalar constant.
 - Express $\vec{AB}$ and $\vec{CD}$ in terms of $\mathbf{a}$, $\mathbf{b}$, and k .
 - If line segment AB is parallel to line segment CD , find all possible values of k .
 - For each value of k found in (b), determine whether $ABDC$ forms a trapezium or a parallelogram.
- In $\triangle ABC$, point D lies on BC such that $BD : DC = 1 : 2$, and point E lies on AC such that $AE : EC = 3 : 1$. The line segments AD and BE intersect at point P .
 - Let $\vec{AP} = k\vec{AD}$ and $\vec{BP} = m\vec{BE}$. Express $\vec{AP}$ in terms of $\vec{AB}$, $\vec{AC}$, and k .
 - By expressing $\vec{AP}$ in terms of $\vec{AB}$, $\vec{AC}$, and m , find the values of k and m .
 - Determine ratios $AP : PD$ and $BP : PE$.
 - If CP extended intersects AB at F , find ratio $AF : FB$.
- Let $ABCD$ be an arbitrary planar quadrilateral. Points M, N, P , and Q are the midpoints of sides AB, BC, CD , and DA respectively.
 - Express $\vec{MN}$ and $\vec{QP}$ in terms of position vectors $\vec{OA}, \vec{OB}, \vec{OC}$, and $\vec{OD}$.
 - Prove that $MNPQ$ is a parallelogram (Varignon's Theorem).
 - Show that the centroid of quadrilateral $MNPQ$ coincides with the centroid of $ABCD$.
- In $\triangle ABC$, let $a = |\vec{BC}|$, $b = |\vec{AC}|$, and $c = |\vec{AB}|$. Point D lies on side BC such that AD is the internal angle bisector of $\angle BAC$.
 - Using unit vectors along $\vec{AB}$ and $\vec{AC}$, show that the position vector of D relative to origin A is:

$$\vec{AD} = \frac{b\vec{AB} + c\vec{AC}}{b + c}$$
 - Deduce the Angle Bisector Theorem: $BD : DC = c : b$.
 - Express $\vec{AD}$ in terms of position vectors $\vec{OA}, \vec{OB}$, and $\vec{OC}$ relative to an arbitrary origin O .
- Let H be the orthocenter of $\triangle ABC$.
 - State the orthogonality conditions between line segments $AH \perp BC$, $BH \perp AC$, and $CH \perp AB$ in terms of scalar products.
 - Prove that $\vec{HA} \cdot \vec{HB} = \vec{HB} \cdot \vec{HC} = \vec{HC} \cdot \vec{HA}$.
 - If $\vec{HA} \cdot \vec{HB} = -12$, evaluate $\vec{HA} \cdot \vec{BC}$.
- Let AB be a diameter of a circle centered at origin O with radius R . P is an arbitrary point on the circle.
 - Express $\vec{PA}$ and $\vec{PB}$ in terms of position vectors $\vec{OP}$ and $\vec{OA}$.
 - Using scalar products, evaluate $\vec{PA} \cdot \vec{PB}$ and deduce Thales' Theorem ($\angle APB = 90^\circ$).
 - If Q is a point inside the circle, determine the sign of $\vec{QA} \cdot \vec{QB}$.

7. Line L passes through point $A(1, 2)$ with direction vector $\mathbf{d} = 3\hat{i} + 4\hat{j}$. Point P has coordinates $(4, 5)$.

- Find vector projection $\vec{p}$ of $\vec{AP}$ onto direction vector $\mathbf{d}$.
- Find the foot of perpendicular N from P onto line L .
- Calculate the perpendicular distance from point P to line L .

8. Consider vectors $\mathbf{u} = a_1\hat{i} + a_2\hat{j}$ and $\mathbf{v} = b_1\hat{i} + b_2\hat{j}$, where $a_1, a_2, b_1, b_2 \in \mathbb{R}$.

- Using the property $(\mathbf{u} \cdot \mathbf{v})^2 \leq |\mathbf{u}|^2 |\mathbf{v}|^2$, prove Cauchy-Schwarz Inequality in 2D:

$$(a_1b_1 + a_2b_2)^2 \leq (a_1^2 + a_2^2)(b_1^2 + b_2^2)$$

- State geometric condition under which equality holds.
- Find maximum value of $3x + 4y$ subject to constraint $x^2 + y^2 = 25$.

9. Let $ABCD$ be a parallelogram.

- Express diagonal vectors $\vec{AC}$ and $\vec{BD}$ in terms of side vectors $\vec{AB}$ and $\vec{AD}$.
- Evaluate scalar product $\vec{AC} \cdot \vec{BD}$.
- Prove that $ABCD$ is a rhombus if and only if its diagonals are perpendicular ($\vec{AC} \perp \vec{BD}$).

10. In $\triangle ABC$, point P lies in the interior such that:

$$2\vec{PA} + 3\vec{PB} + 4\vec{PC} = \vec{0}$$

- Express position vector $\vec{OP}$ in terms of $\vec{OA}$, $\vec{OB}$, and $\vec{OC}$.
- By taking cross products with $\vec{PB}$ and $\vec{PC}$, find ratios of vector areas $|\vec{PA} \times \vec{PB}| : |\vec{PB} \times \vec{PC}| : |\vec{PC} \times \vec{PA}|$.
- Deduce the area ratio $\text{Area}(\triangle PBC) : \text{Area}(\triangle PCA) : \text{Area}(\triangle PAB)$.

11. Consider two unit vectors $\hat{u} = \cos \alpha \hat{i} + \sin \alpha \hat{j}$ and $\hat{v} = \cos \beta \hat{i} + \sin \beta \hat{j}$ on the Cartesian plane.

- Compute scalar product $\hat{u} \cdot \hat{v}$ to derive the compound angle formula for $\cos(\alpha - \beta)$.
- Compute vector product $\hat{u} \times \hat{v}$ to derive the compound angle formula for $\sin(\alpha - \beta)$.
- Evaluate $\cos 15^\circ$ and $\sin 15^\circ$ using vector products with $\alpha = 45^\circ$ and $\beta = 30^\circ$.

12. Let O and H be the circumcenter and orthocenter of $\triangle ABC$ respectively.

- Show that $\vec{OH} = \vec{OA} + \vec{OB} + \vec{OC}$ (Euler line vector formula).
- If G is the centroid of $\triangle ABC$, prove that O, G , and H are collinear.
- Find ratio $OG : GH$.

13. Given vectors $\mathbf{a} = 4\hat{i} - 2\hat{j}$ and $\mathbf{b} = \hat{i} + 3\hat{j}$.

- Express $\mathbf{a}$ in the form $\mathbf{a} = \mathbf{a}_1 + \mathbf{a}_2$, where $\mathbf{a}_1$ is parallel to $\mathbf{b}$ and $\mathbf{a}_2$ is perpendicular to $\mathbf{b}$.
- Calculate magnitudes $|\mathbf{a}_1|$ and $|\mathbf{a}_2|$.
- Verify that $|\mathbf{a}|^2 = |\mathbf{a}_1|^2 + |\mathbf{a}_2|^2$.

14. Let P be an interior point of $\triangle ABC$. Lines AP, BP , and CP intersect opposite sides BC, CA , and AB at D, E , and F respectively.
- (a) Given $\vec{AP} = x\vec{AB} + y\vec{AC}$ ($x, y > 0, x + y < 1$), express ratios $BD : DC, CE : EA$, and $AF : FB$ in terms of x and y .
- (b) Prove Ceva's Theorem using vector ratios:

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1$$

15. In $\triangle ABC$, side lengths opposite to vertices A, B, C are a, b, c respectively. Let I be the incenter of $\triangle ABC$.
- (a) Using the Angle Bisector Theorem, show that the position vector of I relative to origin O is:

$$\vec{OI} = \frac{a\vec{OA} + b\vec{OB} + c\vec{OC}}{a + b + c}$$

- (b) If $A(0, 6)$, $B(-4, 0)$, and $C(4, 0)$, find coordinates of incenter I .
16. Points A and B have coordinates $(2, 3)$ and $(6, 7)$ respectively.
- (a) Find the equation of locus of point $P(x, y)$ satisfying $(\vec{OP} - \vec{OA}) \cdot (\vec{OP} - \vec{OB}) = 0$.
- (b) Identify geometric shape of the locus and state its center and radius.
17. In $\triangle ABC$, M is the midpoint of side BC .
- (a) Express $\vec{AB}$ and $\vec{AC}$ in terms of $\vec{AM}$ and $\vec{MB}$.
- (b) Prove Apollonius' Theorem using scalar products:

$$|\vec{AB}|^2 + |\vec{AC}|^2 = 2(|\vec{AM}|^2 + |\vec{BM}|^2)$$

- (c) If $AB = 6$, $AC = 8$, and $BC = 10$, find length of median AM .
18. Let $ABCD$ be a parallelogram.
- (a) Prove the Parallelogram Law for diagonals:

$$|\vec{AC}|^2 + |\vec{BD}|^2 = 2(|\vec{AB}|^2 + |\vec{BC}|^2)$$

- (b) If $|\vec{AB}| = 5$, $|\vec{BC}| = 7$, and $|\vec{AC}| = 10$, calculate length of diagonal BD .
19. Points $A(t, 2t)$, $B(3, t + 1)$, and $C(1 - t, 4)$ are given on the Cartesian plane, where $t \in \mathbb{R}$.
- (a) Express $\vec{AB} \times \vec{AC}$ in terms of t .
- (b) Find all values of t such that points A, B , and C are collinear.
- (c) For $t = 0$, calculate area of $\triangle ABC$.
20. In isosceles triangle $\triangle ABC$ with $AB = AC$, medians BE and CF are perpendicular to each other.
- (a) Set up position vectors relative to midpoint M of BC , and express $\vec{BE}$ and $\vec{CF}$ in vector form.
- (b) Using $\vec{BE} \cdot \vec{CF} = 0$, determine ratio $AM : BC$.

- (c) Find exact value of $\tan A$.
21. A particle is acted upon by constant forces $\mathbf{F}_1 = 3\hat{i} + 5\hat{j}$ N and $\mathbf{F}_2 = -\hat{i} + 2\hat{j}$ N.
- (a) Find resultant force vector $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2$.
- (b) Calculate displacement vector $\mathbf{d}$ as particle moves from $A(1, -2)$ m to $B(4, 3)$ m.
- (c) Calculate total work done $W = \mathbf{F} \cdot \mathbf{d}$ in Joules.
22. Let vector $\mathbf{v} = x\hat{i} + y\hat{j}$.
- (a) Show that rotating $\mathbf{v}$ counter-clockwise by 90° yields vector $\mathbf{v}^\perp = -y\hat{i} + x\hat{j}$.
- (b) Verify that $\mathbf{v} \cdot \mathbf{v}^\perp = 0$ and $|\mathbf{v}^\perp| = |\mathbf{v}|$.
- (c) Given $\mathbf{w} = 5\hat{i} - 12\hat{j}$, find two unit vectors orthogonal to $\mathbf{w}$.
23. Two particles A and B move on the plane. At time $t \geq 0$, particle A starts at origin $(0, 0)$ with velocity vector $\mathbf{v}_A = 3\hat{i} + 4\hat{j}$, while particle B starts at $(10, 0)$ with velocity vector $\mathbf{v}_B = -\hat{i} + 2\hat{j}$.
- (a) Express position vectors $\vec{OA}(t)$ and $\vec{OB}(t)$ in terms of t .
- (b) Express displacement vector $\vec{AB}(t)$ in component form.
- (c) Find time t at which distance between A and B is minimized, and determine minimum distance.
24. A regular n -gon is inscribed in a unit circle centered at origin O .
- (a) Express position vectors of vertices P_k in polar form for $k = 0, 1, \dots, n-1$.
- (b) Calculate vector area $O\vec{P}_k \times O\vec{P}_{k+1}$ of triangular sector OP_kP_{k+1} .
- (c) Prove that total area of inscribed regular n -gon is:

$$S_n = \frac{n}{2} \sin\left(\frac{2\pi}{n}\right)$$

25. A straight line intersects sides AB, AC , and extended side BC of $\triangle ABC$ at points D, E , and F respectively.
- (a) Express position vectors of D, E , and F using section ratios $\lambda = AD/DB$, $\mu = CE/EA$, and $\nu = BF/FC$.
- (b) Using collinearity condition of D, E , and F , prove Menelaus' Theorem:

$$\frac{AD}{DB} \times \frac{BF}{FC} \times \frac{CE}{EA} = 1$$

26. Point A has coordinates $(3, 8)$. Line L is given vectorially by $\vec{r}(t) = (1 + 2t)\hat{i} + (2 + t)\hat{j}$, where $t \in \mathbb{R}$.
- (a) Find foot of perpendicular N from A onto line L using vector projection.
- (b) Find position vector of reflection point A' of A across line L .
27. Given vectors $\mathbf{a}$ and $\mathbf{b}$ with $|\mathbf{a}| = 3$, $|\mathbf{b}| = 5$, and $|\mathbf{a} + \mathbf{b}| = 7$.
- (a) Evaluate $\mathbf{a} \cdot \mathbf{b}$.
- (b) Evaluate expansion of $(2\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} + 3\mathbf{b})$.

- (c) Calculate magnitude $|2\mathbf{a} - \mathbf{b}|$.
28. Non-parallel vectors are given by $\mathbf{a} = 3\hat{i} + 4\hat{j}$ and $\mathbf{b} = 5\hat{i} + 12\hat{j}$.
- (a) Find unit vectors $\hat{a}$ and $\hat{b}$.
- (b) Show that vector $\mathbf{w} = \hat{a} + \hat{b}$ bisects the angle between $\mathbf{a}$ and $\mathbf{b}$.
- (c) Express $\mathbf{w}$ in component form and find its unit vector.
29. Let $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ be position vectors of vertices of $\triangle ABC$ relative to origin O .
- (a) Express squared side lengths $|\vec{AB}|^2$, $|\vec{BC}|^2$, and $|\vec{CA}|^2$ in terms of dot products of $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.
- (b) Prove that $\triangle ABC$ is equilateral if and only if:
- $$|\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2 - \mathbf{a} \cdot \mathbf{b} - \mathbf{b} \cdot \mathbf{c} - \mathbf{c} \cdot \mathbf{a} = \frac{1}{2} (|\vec{AB}|^2 + |\vec{BC}|^2 + |\vec{CA}|^2)$$
30. In $\triangle ABC$, vertices are given by $A(0, 6)$, $B(-4, 0)$, and $C(4, 0)$.
- (a) Find position vectors of centroid G , orthocenter H , and circumcenter O' relative to origin.
- (b) Verify that O' , G , and H are collinear and find ratio $O'G : GH$.
- (c) Using vector cross products, calculate areas of $\triangle GAB$, $\triangle GBC$, and $\triangle GCA$, and verify they are equal.